

A cAMP-phosphodiesterase controls PKA-dependent differentiation

Gad Shaulsky[†], Danny Fuller and William F. Loomis*

Center for Molecular Genetics, Department of Biology, University of California San Diego, La Jolla, CA 92093, USA

*Corresponding author (e-mail: wloomis@ucsd.edu)

[†]Present address: Department of Molecular and Human Genetics, Baylor College of Medicine, One Baylor Plaza, Houston, TX 77030, USA

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SUMMARY

A cAMP-specific phosphodiesterase was found that is stimulated by binding to the regulatory subunit of cAMP-dependent protein kinase, PKA-R, from either *Dictyostelium* or mammals. The phosphodiesterase is encoded by the *regA* gene of *Dictyostelium*, which was recovered in a mutant screen for strains that sporulate in the absence of signals from prestalk cells. The sequence of RegA predicts that it will function as a member of a two-component system. Genetic analyses indicate that

inhibition of the phosphodiesterase results in an increase in the activity of PKA, which acts at a check point for terminal differentiation. Conserved components known to affect memory, learning and differentiation in flies and vertebrates suggest that a similar circuitry functions in higher eukaryotes.

Key words: *Dictyostelium*, Protein kinase, Two-component system, Sporulation

INTRODUCTION

cAMP dependent protein kinase (PKA) is a central component of several signal transduction pathways in *Dictyostelium*, *Drosophila* and vertebrates (Hammerschmidt et al., 1996; Harwood et al., 1992a,b; Lane and Kalderon, 1993; Parent and Devreotes, 1996). PKA is found as a heterodimer in *Dictyostelium*, with a single catalytic subunit bound to a regulatory subunit, while it is found as a heterotetramer in metazoans where there are two copies of both the catalytic subunit and the regulatory subunit (Leichtling et al., 1984; Mutzel et al., 1987; Taylor et al., 1990). The protein kinase activity of PKA is inhibited as long as the regulatory subunit(s) is bound to the catalytic subunit(s). The regulatory subunit, PKA-R, dissociates from the complex when it binds cAMP at two slightly different sites. The catalytic subunit, PKA-C, is then able to catalyze the phosphorylation of serine or threonine groups in diverse proteins and affect downstream events (Lee, 1991). In *Drosophila* these include inhibition of expression of *decapentaplegic* (*dpp*), *patched* (*ptc*) and *wingless* (*wg*) in pathways responsive to the secreted product of *hedgehog* (Jiang and Struhl, 1995; Li et al., 1995; Pan and Rubin, 1995). PKA also acts in a *hedgehog*-responsive pathway in zebrafish embryos (Hammerschmidt et al., 1996). In addition, PKA is involved in learning and memory processes in both flies and mammals (Abel et al., 1997; Skoulakis et al., 1993).

In *Dictyostelium*, PKA is the major effector of cAMP as a second messenger. The essential role of PKA is clearly demonstrated by the failure of cells lacking this activity to even aggregate (Mann and Firtel, 1991; Mann et al., 1992; Harwood et al., 1992a). Mutations that eliminate cAMP binding of PKA-R result in a dominant negative form of the protein (PKA-R_m),

which binds and inactivates the catalytic subunit, but fails to dissociate in the presence of cAMP. Strains expressing PKA-R_m under cell type-specific promoters exhibit a variety of developmental defects. Prespore-specific expression of PKA-R_m results in reduced sporulation and increased sensitivity to ammonia (Hopper et al., 1993), while prestalk-specific expression of PKA-R_m results in developmental arrest at the first finger stage and a cooperative defect in sporulation (Harwood et al., 1992b; Shaulsky et al., 1995). Conversely, overexpression of PKA-C results in precocious sporulation as well as sporulation under conditions that preclude signalling from prestalk cells (Anjard et al., 1992; Mann et al., 1992). Growth and development can proceed without accumulation of cAMP as long as PKA is constitutively active, either as the result of mutations in *pkaR*, the gene encoding the regulatory subunit, or overexpressing *pkaC*, the gene encoding the catalytic subunit (Abe and Yanagisawa, 1983; Simon et al., 1992; Wang and Kuspa, 1997). Cells of these strains aggregate precociously and differentiate into stalk cells and spores at least 6 hours sooner than wild-type cells. It appears that progression through the developmental stages is normally controlled by the level of PKA activity at various check points that are by-passed in strains with constitutive PKA activity.

We recently isolated a mutant with properties reminiscent of *pkaR*⁻ mutants (Shaulsky et al., 1996). The sequence of the newly affected gene, *regA*, suggested that it encodes a cyclic nucleotide phosphodiesterase (PDE) that might reduce the cAMP available to PKA-R. Mutations inactivating such a PDE would be expected to have constitutive PKA activity similar to that in *pkaR*⁻ mutants. We have now measured the cAMP-PDE activity of purified RegA and found that it is stimulated when measured in the presence of purified PKA-R.

The primary sequence of *regA* also indicated that the N-terminal domain of its product might function as a response regulator in a two-component system. Two-component systems consisting of a protein histidine kinase and a response regulator have been found to function in the response of a variety of cells to environmental signals (Alex and Simon, 1994; Appleby et al., 1996; Maeda et al., 1994; Posas et al., 1996; Loomis et al., 1997). Such a system has recently been implicated in the development of *Dictyostelium discoideum* (Wang et al., 1996). Inactivation of the histidine kinase gene *dhkA* by insertional mutagenesis results in the formation of fruiting bodies with aberrantly long stalks and very few spores. Conversely, *regA*⁻ mutants make fruiting bodies with short stalks but essentially the normal number of spores.

The *regA* gene was isolated as a second site suppressor of a mutation in the prestalk gene, *tagB*, which is essential for generating the signal for sporulation (Shaulsky et al., 1996). Cells in which either of two adjacent genes, *tagB* or *tagC*, is disrupted have a cell autonomous defect in prestalk differentiation and fail to make spores (Shaulsky et al., 1995). However, when these cells are allowed to develop together with wild-type cells, the *tag* mutants sporulate normally. The *tagB* and *tagC* genes encode proteins that are 90% identical and are expressed uniquely in prestalk cells (Shaulsky et al., 1995; Shaulsky and Loomis, 1996). They each encode proteins with two domains, one homologous to serine proteases and the other homologous to the ABC family of ATP-driven membrane transporters (Shaulsky et al., 1995).

Mutations in *tagB* or in *tagC* compromise the ability of prestalk cells to produce the signal that initiates encapsulation of prespore cells. RegA appears to be a component of the signal transduction mechanism that activates PKA and triggers terminal differentiation in prespore cells.

MATERIALS AND METHODS

Cells, growth, transformation and development

Cell growth and maintenance in liquid suspension or on SM nutrient agar in association with bacteria were as described (Shaulsky and Loomis, 1993). *Dictyostelium discoideum* strains used in this work were: wild type AX4 cells (Knecht et al., 1986), *tagB* null mutants (Shaulsky et al., 1995), *regA* null mutants (Shaulsky et al., 1996), TL1 (AX4, *cotB::lacZ*) and TL6 (AX4, *ecmA::lacZ*) (Fosnaugh and Loomis, 1993).

Gene disruption by homologous recombination and selection for blasticidin resistance were used for *pkaR* and *regA*, while selection for uracil autotrophy was used to disrupt *tagB* (Kuspa and Loomis, 1992; Shaulsky et al., 1996). Knockout vectors for disruption of *regA* were as described (Shaulsky et al., 1996), while the knockout vector for disruption of *pkaR* was a gift from A. Kuspa. Transformation with the expression vector for *pspA::pkaR_m* (Hopper et al., 1993), a gift from J. Williams, and selection for G418 resistance and maintenance of transformed cell lines were as described (Shaulsky and Loomis, 1993). The molecular genotypes of all of the transformed cell lines were confirmed by Southern blot analysis as described (Shaulsky and Loomis, 1993).

Synchronous development of cells on nitrocellulose filters, slug migration on Nobel agar plates and sporulation assays were performed as described (Shaulsky et al., 1996; Shaulsky and Loomis, 1993). X-gal staining of *lacZ*-expressing cells and counterstaining with eosine Y were as described (Shaulsky et al., 1995).

Sporogenous assay

Sporulation of *Dictyostelium* cells at low density in submerged cultures was modified from the method of Kay and Trevan (1981). Exponentially growing cells were washed twice in spore assay buffer (10 mM MES, pH 6.5, 10 mM NaCl, 10 mM KCl, 1 mM CaCl₂, 1 mM MgSO₄ and 5 mM cAMP) and plated at a density of 2×10⁴ cells per cm² in 2 ml spore assay buffer in 6-well tissue culture plates (Costar, Cambridge, MA). Plates were incubated at 22°C in a humid chamber for 24–36 hours. Detergent (0.3% cernulol) was added to the wells, the cells were collected and the number of visible spores was determined by phase microscopy. Samples were diluted and plated on SM nutrient agar in association with bacteria to determine the number of viable spores.

Bacterial expression of RegA

A *Sall*-*HindIII* DNA fragment consisting of base pairs 48–2343 of the open reading frame of *regA* cDNA (Shaulsky et al., 1996) was cloned into pQE10 (Qiagen Inc., Chatsworth, CA). The resulting plasmid was transformed into *E. coli* M15[pREP] (Qiagen Inc., Chatsworth, CA). Bacterial cells were exponentially grown in Super medium (25 g/l bacto-tryptone, 15 g/l bacto-yeast extract and 5 g/l NaCl) to OD₆₀₀=0.8. Expression of the hexahistidine fusion RegA protein was induced with 2 mM IPTG for 5 hours. Purification of the native protein on TALON metal affinity resin (Clontech, Palo Alto, CA) was performed according to the manufacturer's recommended procedure and the protein was eluted in a buffer containing 100 mM imidazole. Fractions were assayed for phosphodiesterase activity and those with activity were stored at 4°C with 0.1 mM PMSF. The activity was stable under these conditions for 2 weeks. The 100 kDa recombinant RegA protein accounted for about 50% of the total protein in the 100 mM imidazole eluent, as estimated from silver-stained gels following electrophoresis on an 8% polyacrylamide SDS gel. No phosphodiesterase activity or band at 100 kDa was seen in lysates from untransformed *E. coli* M15[pREP] treated in a similar manner. Further purification of RegA for labeling with acetyl phosphate was achieved by gel filtration through a spin-column of G50 Sephadex (Pharmacia, Uppsala, Sweden) equilibrated with 20 mM Tris HCl, pH 7.5, and 10 mM MgCl₂. For cAMP phosphodiesterase activity, the 100 mM imidazole eluent was purified by FPLC gel filtration through Superose 12 (Pharmacia, Uppsala, Sweden) in 50 mM Tris HCl, 0.1 mM PMSF. Fractions with the highest PDE activity were analyzed. Protein concentration was determined using the BIO-RAD protein assay reagent with BSA as standard.

Purified PKA-R protein

PKA-R proteins were expressed as hexahistidine fusions in *E. coli* and purified by affinity chromatography. The *Dictyostelium* PKA-R protein was stripped of cAMP according to Buechler et al. (1993) and was gift from R. Biondi and M. Veron (Pasteur Institute, Paris). Wild-type bovine PKA-R^{I_a} and PKA-R^{I_a}Δ1–91 (Buechler and Taylor, 1991; Herberg et al., 1994; Saraswat et al., 1988) were a gift from L. Huang and S. Taylor (UCSD, La Jolla, CA).

Cyclic nucleotide phosphodiesterase assay

Phosphodiesterase (PDE) assays were modified from Tsang and Coukell (1977). Known amounts of protein were mixed in a final volume of 50 µl with 20 mM Tris HCl, pH 7.5, 10 mM MgCl₂ and a mixture of 1 µCi [³H]-labeled and the respective unlabeled cyclic nucleotide. Reactions were mixed on wet ice and initiated by incubation at 30°C for 30–180 minutes. The reactions were terminated by addition of 5% TCA (w/v), precipitated proteins were removed by centrifugation for 5 minutes at top speed in an Eppendorf centrifuge and the supernatants were concentrated by evaporation under reduced pressure to about 5 µl. Samples were spotted on Whatman No. 1 paper, air-dried and resolved by ascending paper chromatography in 95% ethanol, 1 M ammonium acetate, pH 7.5 (75:30 by volume). The

regions corresponding to the cyclic nucleotide substrate ($R_f=0.5-0.7$) and to the 5'-nucleotide monophosphate ($R_f=0-0.1$) were cut out and the radioactivity determined by liquid scintillation. All of the measurements were performed in duplicate or in triplicate and the data are representative of 2-5 independent experiments. K_m was determined by the double-reciprocal method of Leinweaver and Burk. To determine the pH optimum, Tris HCl, pH 7.5 in the reaction mix was replaced with 20 mM sodium phosphate buffers ranging from pH 5.7 to pH 8.5 in 0.5 increments and the pH optimum was found to be 7.5. Accumulation of the product was linear with time and the amount of RegA protein added (data not shown). Radioactive [^3H]cAMP (27 Ci/mmol), and [^3H]cGMP (9.3 Ci/mmol) were purchased from DuPont NEN, Boston, MA.

Stimulation of RegA PDE activity by PKA-R

RegA (1.2 μg) was incubated at 4°C for 15-17 hours alone or with 6 μg of the respective PKA-R protein in a final volume of 150 μl containing 40 mM Tris HCl, pH 7.5 and 10 mM MgCl_2 . The reactions were divided into six equal portions, mixed in duplicates with the indicated concentrations of [^3H]cAMP, and PDE activity determined. The experiments were repeated three times with essentially identical results.

Binding of RegA to PKA-R

Dictyostelium PKA-R (1 μg) was incubated at 4°C for 15-17 hours alone or with 8 μg of RegA protein or BSA in a final volume of 100 μl containing 20 mM Tris HCl, pH 7.5 and 10 mM MgCl_2 . The proteins were resolved by gel filtration on an FPLC Superose 12 column in 50 mM Tris HCl, 0.1 mM PMSF. Fractions (1 ml) were collected and samples of 0.5 ml were loaded on a nitrocellulose membrane using a vacuum dot blot apparatus (Schleicher and Schuell). Twofold serial dilutions of purified *Dictyostelium* PKA-R were loaded on the same nitrocellulose membrane as a quantitation standard. The membrane was subjected to western blot analysis with a rabbit polyclonal antiserum against PKA-R (a gift from M. Veron) as described (Shaulsky et al., 1995). The staining intensity of each sample was compared to the standard to determine the amount of PKA-R protein. The experiment was repeated twice with essentially identical results.

Phosphorylation by acetyl phosphate

Purified RegA protein was incubated in a final volume of 30 μl containing 50 mM Tris HCl, pH 7.0, 5 mM MgCl_2 , 1 mM DTT, 0.2 mg/ml protein and 10 mM [^{32}P]acetyl phosphate (50 mCi/mmol), custom synthesized by DuPont NEN, Boston, MA. EDTA (100 mM) was included in the reaction mix as indicated. Reactions were incubated for 20 minutes at 25°C, transferred to ice and mixed immediately with SDS-PAGE sample buffer (3% β -mercaptoethanol, 3% SDS, 0.3% bromophenol blue, 10% glycerol). Samples were loaded without boiling and separated by electrophoresis on an 8% polyacrylamide SDS gel at 4°C with pre-stained molecular markers (BRL); the gel was dried under reduced pressure and exposed to X-ray film for autoradiography. The gel was then soaked in 3 M KOH for 1 hour at room temperature, rinsed in double-distilled water for 30 minutes, dried again and exposed to X-ray film for autoradiography. The pre-stained molecular markers were retained on the gel throughout the procedure.

RESULTS

RegA is a cAMP-specific phosphodiesterase

The primary sequence of *regA* suggested that the encoded protein might function as a cyclic nucleotide PDE as well as a response regulator in a two-component system (Shaulsky et al., 1996). To test these predictions we cloned a cDNA fragment

that consisted of most of the open reading frame of *regA* into a bacterial expression system. RegA was expressed as a hexahistidine fusion protein in *E. coli*, purified and used for biochemical studies.

Cyclic nucleotide phosphodiesterases catalyze the hydrolysis of 3', 5' cyclic nucleotide monophosphates into the respective 5' nucleotide monophosphates. They can use cAMP, cGMP or both as substrates, and various enzyme activities on one cyclic nucleotide can be affected by the presence of another cyclic nucleotide (Beavo, 1990). To test the substrate specificity of RegA, purified protein was incubated with [^3H]cAMP with or without unlabeled cGMP or with [^3H]cGMP alone. PDE activity was measured by paper chromatography followed by liquid scintillation, and the results showed that RegA is a PDE – active with cAMP but inactive towards cGMP (Fig. 1A). The activity with [^3H]cAMP was slightly stimulated by the presence of cGMP at concentrations up to 10 mM (Fig. 1B). Maximal cAMP-PDE activity was measured at pH 7.5 and the K_m for cAMP was 4 μM (data not shown).

Similarities between *regA*⁻ and *pkaR*⁻ mutants

The phenotypes resulting from null mutations in *regA* were reminiscent of those seen in *rdeC* mutants that are rapid developers due to lesions in the *pkaR* gene (Abe and Yanagisawa, 1983; Simon et al., 1992). In order to study the similarities between *regA*⁻ and *pkaR*⁻ mutants, we transformed wild-type AX4 cells with a construct in which the blasticidin resistance gene was inserted into the coding region of *pkaR*, and selected for strains that had integrated the disrupted gene by homologous recombination. The phenotypes of our *pkaR* null strains were indistinguishable from those described for *rdeC* mutants (Abe and Yanagisawa, 1983) and similar to those of *regA* null mutant strains.

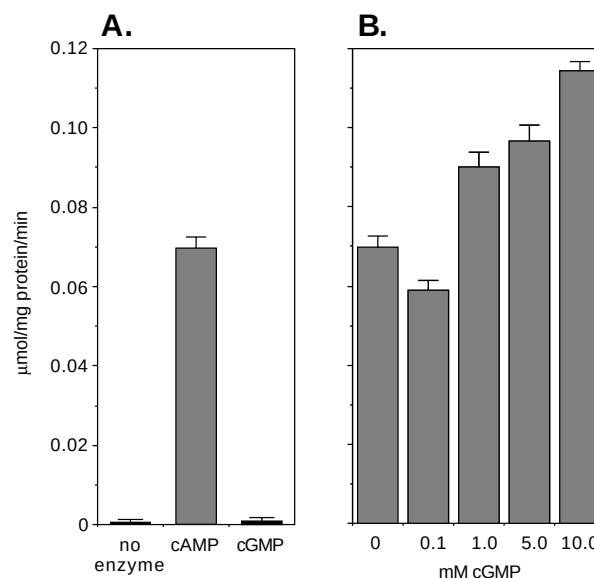


Fig. 1. RegA is a cAMP-specific phosphodiesterase. (A) PDE activity of RegA was determined with 100 μM [^3H]cAMP or with 100 μM [^3H]cGMP. A control with no enzyme was done with 100 μM [^3H]cAMP and subtracted from subsequent measurements. (B) PDE activity of RegA was measured with 100 μM [^3H]cAMP in the presence of the indicated concentrations of unlabeled cGMP.

In comparison to wild-type strains, both the *pkaR* null mutants and *regA* null mutants were precocious when developed synchronously on filters (Fig. 2A). The time of appearance of viable spores was determined by collecting samples at various times in detergent that lyses unencapsulated cells, spreading samples on SM nutrient agar in association with bacteria, and counting the number of plaques that arose after a week in the resulting bacterial lawns. In developing wild-type populations spores were first found after 20 hours and reached a maximum by 28 hours, but in developing populations of both the *regA* null mutants and *pkaR* null mutants spores appeared at 16 hours and reached maximal levels by 20 hours. Note that *regA* null spores were stable whereas *pkaR* null spores rapidly lost viability after 20 hours of development (Fig. 2A). Spore instability is a characteristic found in other rapid-developing strains (Osherov et al., 1997; Sadiq and Town, 1991), suggesting that premature encapsulation sometimes fails to generate fully differentiated spores.

When spores were collected from wild type, *regA*⁻ or *pkaR*⁻ strains after 48 hours of development, treated with detergent and plated on SM agar, 80% of the visible spores of the wild type formed plaques after 3 days and the remainder formed visible plaques the next day. However, less than half of the visible spores from either *pkaR* null mutants or from *regA* null mutants formed plaques by 4 days, while the remainder formed plaques gradually over the following week. Once a plaque appeared, it grew at the wild-type rate (data not shown). It appears that spore germination is impaired in both the *pkaR*⁻ mutants and the *regA*⁻ mutants. Our results are consistent with the findings of van Es et al. (1996), who showed the involvement of cAMP and PKA-C in regulation of germination by the osmo-sensitive adenylyl cyclase *AcgA*. It is likely that *RegA* plays a role in that pathway as well.

Mutations in *pkaR* are known to result in aberrant fruiting body morphology, due to compromised stalk formation, without affecting cell type proportioning (Abe and Yanagisawa, 1983). To examine the effect of *regA* on cell type proportioning and overall morphology we generated *regA* null strains that carry the prespore marker *cotB::lacZ* or the prestalk marker *ecmA::lacZ*. Cells of these strains were developed on filters, fixed and stained with X-gal to detect β -galactosidase activity in the respective cell types. At 16 hours of development the prespore cells of the *regA*⁻ *cotB::lacZ* strain were in the posterior, as expected, and the ratio of prespore to prestalk cells appeared normal (Fig. 2B). The overall morphology of the structures seen at this time was not significantly different from those seen in wild-type strains but spores had already appeared (Fig. 2B). Thus, precocious sporulation of *regA* null cells does not result from an overall acceleration of the developmental process, but rather from uncoupling of sporulation from normal morphogenesis. Subsequent morphogenesis was abnormal, probably as the result of the altered properties of the encapsulated spores. Fruiting bodies were formed in which the mass of spores was at the bottom of a short stalk rather than at the apex as seen in wild-type strains. When *regA*⁻ mutant cells carrying the prestalk specific reporter construct, *ecmA::lacZ*, were stained after 24 hours of development, the vacuolized cells in the short stalk were clearly blue (Fig. 2C). Development of these strains as chimeras following mixing with vegetative cells of wild-type strains at the start of

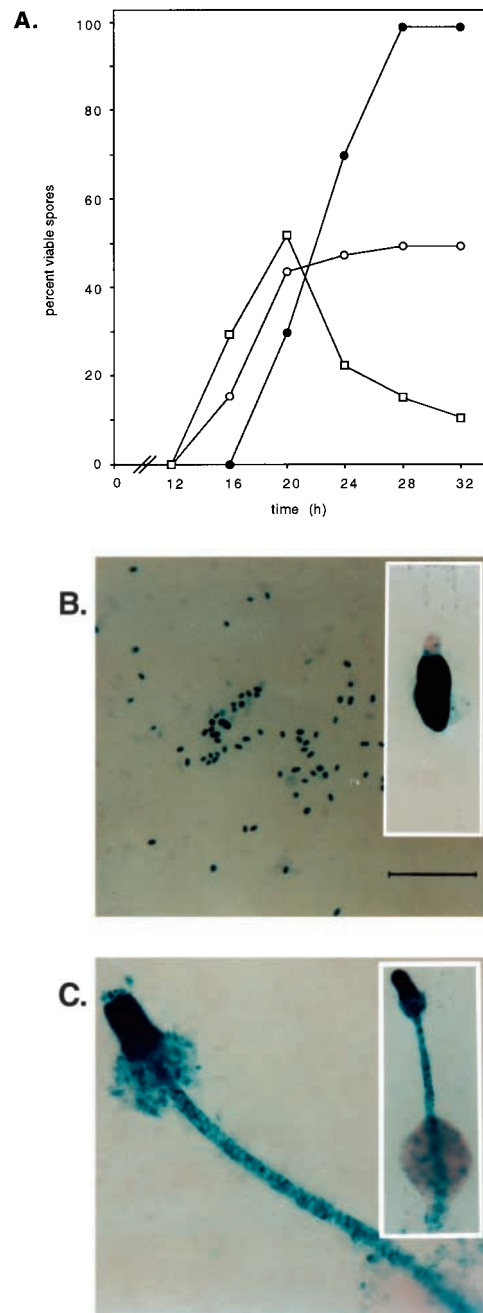


Fig. 2. Precocious sporulation and developmental morphology of *regA* null mutants. (A) Cells were developed synchronously on nitrocellulose filters. At the indicated time points, cells were treated with detergent, washed and plated on nutrient agar in association with bacteria. The number of viable spores is presented as a fraction of the original number of cells developed. Closed circles, wild-type AX4 cells; open circles, *regA* null mutants; open squares, *pkaR* null mutants. (B) *regA* null mutants carrying the prespore reporter gene *cotB::lacZ* were developed on filters for 16 hours. Cells were fixed, stained with X-gal (blue) and counterstained with eosine Y (pink). Multicellular structures were dispersed to show spores in the main panel or photographed intact in the insert. Bar, 0.1 mm (0.2 mm for insert). (C) *regA* null mutants carrying the prestalk reporter gene *ecmA::lacZ* were developed on filters for 24 hours. Cells were fixed, stained with X-gal (blue) and counterstained with eosine Y (pink). Vacuolized stalk cells are shown in the main panel and a whole structure is shown in the insert. Magnifications as in B.

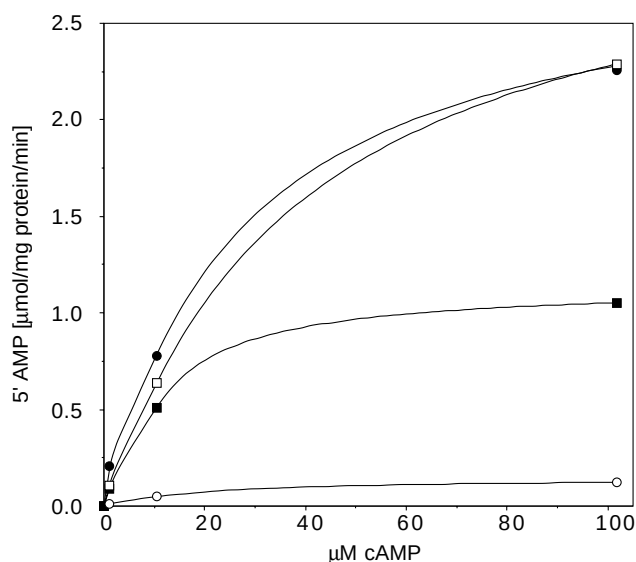


Fig. 3. Stimulation of RegA phosphodiesterase activity by various PKA-R proteins. RegA protein was incubated overnight at 4°C with 0.5 μM PKA-R protein in the absence of cAMP. PDE activity was measured with [³H]cAMP as indicated. Open circles, no PKA-R; closed circles, wild-type *Dictyostelium* PKA-R; closed squares, wild type bovine PKA-R^{1α}; open squares, bovine PKA-R^{1α} Δ1-91 lacking the dimerization domain.

development showed that the *regA* null phenotype of rapid sporulation was cell autonomous (data not shown).

Constitutive PKA activity results in cells that are able to form spores in the absence of morphogenesis or cell-cell contact when incubated under a layer of buffer in Petri dishes (Anjard et al., 1992). Wild-type cells make very few spores under these conditions. To determine whether *regA* null cells were sporogenous, they were incubated at low cell density under buffer for 36 hours. The cells were then treated with detergent and the number of visible spores was determined. Spores were plated on SM nutrient agar in association with bacteria to determine their viability. As shown in Table 1, both *regA* null mutants and *pkaR* null mutants formed a considerable number of spores whereas wild-type cells failed to sporulate efficiently under the same conditions. The viability of *regA* null spores formed under these conditions was higher than that of the *pkaR* null spores (Table 1), consistent with the relative viability of spores of these strains developed on filters (Fig. 2A). In conclusion, *regA*[−] mutants are similar to *pkaR*[−] mutants in that they are rapid developers, they have compromised stalk development, they are sporogenous and their spore germination is compromised.

Association between RegA and PKA-R proteins

The phenotypic similarities between *regA* null mutants and *pkaR* null mutants led us to consider the possibility that the PDE activity of RegA might affect PKA by reducing the levels of cAMP within cells. Moreover, a direct association of RegA and PKA-R would concentrate the phosphodiesterase activity at the site of action. Therefore, we tested the PDE activity of RegA after incubation in the presence or absence of PKA-R. Following purification over a Superose column, RegA was incubated overnight with or without purified *Dictyostelium* PKA-R at 4°C in the absence of cAMP and the PDE activity

Table 1. *regA* null mutants and *pkaR* null mutants are sporogenous

Genotype	Visible spores	Viable spores
Wild type	2.6%	0.4%
<i>regA</i> null	40.0%	10.1%
<i>pkaR</i> null	66.7%	1.5%

The number of spores is presented as a fraction of the original number of cells developed.

was then assayed at 30°C with [³H]cAMP at various concentrations. Purified RegA is stable at 4°C in the absence of added proteins and is routinely stored this way. We found that PKA-R stimulated the PDE activity of RegA at least 18-fold without significantly affecting the *K_m* for cAMP (Fig. 3). Strong stimulation can be seen at 10 μM cAMP, the concentration found within developing cells (Gersich et al., 1977). The enzyme activity was linear with respect to time over the period of the assay in both the presence and absence of purified PKA-R subunit. The phosphodiesterase activity of RegA was unaffected by pre-incubation with 0.5 μM bovine serum albumin (data not shown). Since our preparation of purified PKA-R had no measurable PDE activity when assayed alone, it appears that the cAMP-PDE activity of RegA is dramatically stimulated by an interaction with PKA-R. Consistent with the need to form complexes in dilute solution, we found that the rate of activation of RegA activity was dependent on the concentration of added PKA-R; half-maximal stimulation was found within an hour at 5 μM PKA-R while it was about 2 hours at 0.5 μM PKA-R (data not shown).

Primary sequence analysis indicates that *Dictyostelium* PKA-R is a member of the type I family of mammalian regulatory subunits (Mutzel et al., 1987; Taylor et al., 1990). It was therefore interesting to test the effect of bovine PKA-R^{1α} on RegA PDE activity. As shown in Fig. 3, bovine PKA-R^{1α} protein increased the PDE activity by about eightfold (closed squares), indicating that mammalian PKA-R can also effectively associate with this cAMP-PDE, although it stimulates the activity somewhat less than *Dictyostelium* PKA-R. One of the main differences between the *Dictyostelium* PKA-R and mammalian PKA-R^{1α} proteins is the presence of a dimerization domain in the amino terminus of the latter (Mutzel et al., 1987). We therefore tested a modified bovine PKA-R^{1α} that lacks the dimerization domain (Herberg et al., 1994) and found that it was able to stimulate RegA to the same extent as the *Dictyostelium* PKA-R (Fig. 3).

The results presented in Fig. 3 indicate an effective interaction between the RegA and PKA-R proteins. To directly test if RegA can bind *Dictyostelium* PKA-R we incubated the purified proteins separately or together overnight at 4°C, resolved them by gel filtration and measured the amount of PKA-R in the various fractions by western dot-blots with an antibody against *Dictyostelium* PKA-R. Most of the PKA-R protein was eluted in fractions 13-15, consistent with its expected size (Fig. 4, open symbols). After pre-incubation with RegA protein, the elution profile of PKA-R changed dramatically (Fig. 4, closed symbols). Most of the PKA-R protein eluted in fraction 11, ahead of the monomeric PKA-R. Under the same conditions, PDE activity eluted in fraction 12 when purified RegA was run on its own, but eluted in fractions

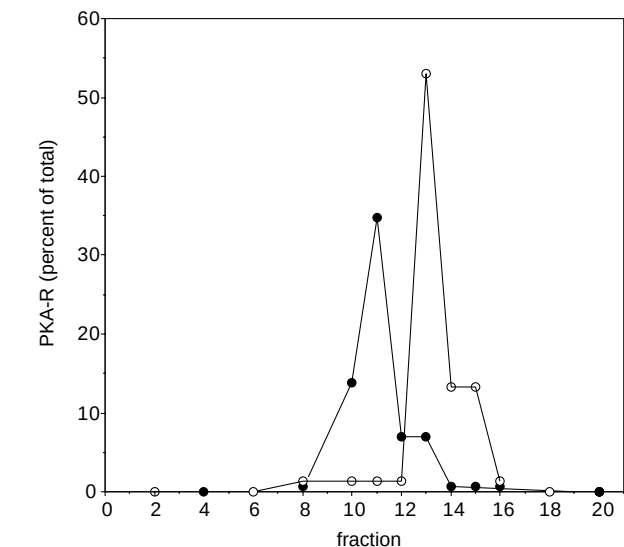


Fig. 4. Binding of PKA-R and RegA. *Dictyostelium* PKA-R was incubated overnight at 4°C with RegA protein. The proteins were resolved by gel filtration, 1 ml fractions were collected and the amount of PKA-R protein was determined by western dot-blot analysis. The data are presented as the proportion in each fraction of the total protein eluted after background subtraction. Open circles, PKA-R alone; closed circles, PKA-R after pre-incubation with excess RegA protein.

10-11 when pre-incubated with excess PKA-R. Preincubation with bovine serum albumin did not affect the elution profile of either PKA-R or RegA (data not shown). The precise mechanism of interaction between PKA-R and RegA is currently under investigation, but the results in Figs 3 and 4 clearly demonstrate a physical association between the two proteins that stimulates the cAMP-PDE activity of RegA.

A genetic pathway that connects PKA to an extracellular signaling system

regA was originally discovered as a genetic suppressor of *tagB* null mutants (Shaulsky et al., 1996). *tagB* null mutants suffer from a cell-autonomous defect that blocks prestalk cell differentiation and affects prespore differentiation in a cooperative manner (Shaulsky et al., 1995). Since *regA* null mutations can suppress the inability of *tagB* null cells to sporulate we suggested that *regA* might be involved in the prespore specific signal transduction mechanism that integrates the *tagB*-dependent signal produced by prestalk cells (Shaulsky et al., 1996). The data shown above that link *regA* to *pkaR* led us to examine if a *pkaR* null mutation would suppress the *tagB* null mutation as *regA* did. The results in Table 2 show that the double mutant *tagB*[−] *pkaR*[−] formed 14% spores after development on filters, a vast increase compared to the parental *tagB*[−] strain. Interestingly, these mutants did not sporulate when the cells were grown in association with bacteria on SM-nutrient agar, explaining our failure to isolate them in the screen for suppressors of *tagB* (Shaulsky et al., 1996). These results support the notion that *tagB*, *regA* and *pkaR* are components of the same signal transduction pathway and that *regA* and *pkaR* function downstream of *tagB*. Inactivation of *regA* would result in constitutively dissociated PKA-R since it would be bound to cAMP and unable to inhibit PKA-C.

Table 2. Epistatic relationships in the sporulation regulation pathway

Genotype	Viable spores
Wild type	100%
<i>tagB</i> [−]	<10 ^{−10} ^a
<i>tagB</i> [−] <i>regA</i> [−]	30% ^a
<i>tagB</i> [−] <i>pkaR</i> [−]	14% ^b
<i>regA</i> [−] <i>pspA</i> :: <i>R_m</i>	1% ^c
<i>regA</i> [−]	50% ^a
<i>pkaR</i> [−]	52%
<i>pspA</i> :: <i>R_m</i>	2% ^c

The number of spores is presented as a fraction of the original number of cells developed.

^aShaulsky et al. (1996)

^bSporulation only on filters

^cAfter slug migration

In order to determine the epistatic relationship between *regA* and *pkaR* in this pathway we transformed *regA* null mutants with a dominant negative allele of *pkaR* under regulation of the *pspA* promoter, a prespore specific promoter that is not dependent on the activity of PKA (Hopper et al., 1993, 1995). As shown in Table 2, *regA* null cells carrying the *pspA*::*pkaR_m* expression vector showed dramatically reduced sporulation efficiency. This result indicates that *pkaR* functions downstream of *regA* in a *tagB*-dependent signal transduction pathway.

Phosphorylation of RegA

In two-component systems a phosphate is transferred from the histidine kinase, either directly or indirectly, to a conserved aspartic acid residue in the D motif of the response regulator. Another characteristic of response regulators is their ability to autophosphorylate the conserved aspartate residue using small acyl-phosphate compounds as phosphate donors, in a reaction dependent on magnesium ions and therefore sensitive to EDTA (Lukat et al., 1992). One of the characteristics of aspartyl phosphates is their rapid hydrolysis under alkali conditions. To test the response-regulator nature of RegA we incubated the protein with [³²P]acetyl phosphate in the presence or absence of EDTA, and resolved the products by polyacrylamide SDS gel electrophoresis. Autoradiography of the resulting gels showed that the RegA protein was labeled and that this phosphorylation was inhibited by EDTA (Fig. 5). Moreover, labeling intensity was directly proportional to the reaction time (data not shown). The same gel was treated with alkali and

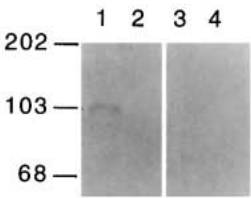
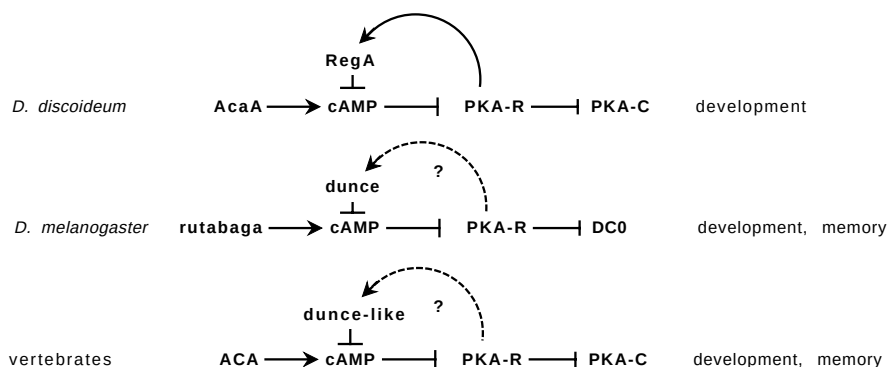


Fig. 5. Phosphorylation of RegA by acetyl phosphate. RegA protein was incubated at 25°C with [³²P]acetyl phosphate for 20 minutes before being electrophoretically resolved on a 8% polyacrylamide SDS gel that was dried and exposed to X-ray film (lanes 1, 2). The gel was then treated with alkali, dried and re-exposed (lanes 3, 4). EDTA (100 mM) was added to the reactions in lanes 2 and 4. The positions of molecular mass (kDa) markers are indicated.

Fig. 6. Conserved components in the PKA regulation circuitry. Regulation of PKA catalytic activity is essential for efficient spore differentiation in *Dictyostelium*. cAMP, which is synthesized by adenylyl cyclase (ACA), can activate PKA-C by binding to the regulatory subunit PKA-R. The response-regulator-PDE encoded by *regA* activates PKA-R such that it can inhibit PKA-C by removal of cAMP. The interaction between PKA-R and RegA suggests that some of the components in the circuitry may be closely associated.

Drosophila homologues of ACA (rutabaga), RegA (dunce) and PKA-C (DCO) are involved in embryogenesis, learning and memory. A

close homologue of the *dunce*-PDE gene is found in the rat brain; moreover, modification of PKA-R activity in the brains of mice results in memory defects, raising the possibility that the proteins encoded by *dunce* and *dunce-like* genes may directly interact with PKA-R.



exposed for a longer time to compensate for the decay of ^{32}P . The results in Fig. 5 show that all the radioactive material was removed by this procedure, consistent with the alkali-labile nature of aspartyl phosphate. The genetic analyses predict that phosphorylation of RegA should result in inhibition of its cAMP-PDE activity or prevent its association with PKA-R. We could not test these predictions because incubation of RegA with acetyl-phosphate resulted in phosphorylation of only a small fraction of the total protein. Therefore, the present results only support the homology-based suggestion that RegA is a response regulator protein in a two-component system.

DISCUSSION

cAMP is a ubiquitous second messenger that is used to adapt cellular physiology in response to environmental and developmental signals (Robinson et al., 1971). When the level of cAMP increases, it binds to the regulatory subunit of PKA, resulting in the release and activation of the catalytic subunit such that it can phosphorylate downstream proteins (Lee, 1991; Meinkoth et al., 1993). Activation of PKA is thought to be a rapidly reversible process as the result of re-association of the subunits when the level of cAMP drops once again. However, the free regulatory subunit binds cAMP so avidly and the rate of cAMP release from PKA-R is so slow that it has not been apparent how the cAMP binding sites are cleared within the cell (Corbin et al., 1978; de Gunzburg and Veron, 1982). Moreover, it has been reported that PKA can be activated without subunit dissociation and that the enzyme responds to a flux of cAMP better than to a fixed level of cAMP (Yang et al., 1995; Leiser et al., 1986). These observations make it difficult to explain the reversible stimulation of PKA activity.

The intimate association of a cAMP-PDE with the R subunit and the subsequent stimulation of its activity helps to account for the major role that PKA plays in mediating the effects of rapidly changing levels of cAMP. We have found that the RegA protein of *Dictyostelium* is able to form a complex with the R subunit and is stimulated by PKA-R subunits from either *Dictyostelium* or from bovine origin. Thus, the association appears to be a general property of these proteins that has been conserved since the time before the radiation of metazoans. RegA may be unique in its ability to be stimulated by PKA-R, since at least one other PDE activity was reduced when

measured on cAMP bound to PKA-R (Corbin et al., 1978), but other genetic pathways that regulate PKA activity in flies and in vertebrates suggest that similar PDEs may well exist (Fig. 6). A search for the mammalian counterpart of *regA* is now underway.

Mutations in any of the *Drosophila* genes *rutabaga*, *dunce* or *DCO* result in flies that are learning impaired and have additional developmental defects (Dudai et al., 1976; Livingstone et al., 1984; Skoulakis et al., 1993; Bellen et al., 1987; Bellen and Kiger, 1987, 1988). *rutabaga* encodes adenylyl cyclase and *dunce* encodes a cAMP-PDE, so mutations in these genes would be expected to affect cAMP levels within the cells (Davis and Kiger, 1981; Levin et al., 1992). Since *DCO* encodes PKA-C, it seems that the memory defects in these mutant flies result from loss of cAMP control of PKA activity (Drain et al., 1991; Kalderon and Rubin, 1988; Skoulakis et al., 1993). A role for PKA in higher cognitive processes such as long-term potentiation and memory in transgenic mice has been directly demonstrated by expressing a modified gene, R(AB), which encodes a PKA-R protein that is unable to bind cAMP at either of its two sites as the result of site-directed mutations (Abel et al., 1997). Expression of this dominant-negative gene under the control of the Ca^{2+} /calmodulin protein kinase IIa regulatory region in the hippocampus results in mice with defects in spatial and long-term memory. Likewise, expression of a dominant negative allele of the *Dictyostelium* PKA-R that does not bind cAMP blocks prestalk and prespore differentiations when expressed in the respective cells (Harwood et al., 1992b; Hopper et al., 1993; Zhukovskaya et al., 1996). Furthermore, the block to sporulation resulting from expression of *pspA::pkaR_m* is epistatic to the constitutive sporulation resulting from mutations in *regA*. Thus, the cAMP-PDE encoded by *regA* appears to be mainly directed at the control of PKA activity. Apparently, sporulation in *Dictyostelium* is regulated by the same components that control differentiation and memory in flies and in mice (Fig. 6).

The primary sequence of RegA led us to consider that it might be a member of a two-component system that is regulated by the activity of a histidine kinase. The three aspartate residues that bind magnesium and position the acceptor aspartate for phosphorylation in the bacterial response regulator CheY (Stock et al., 1993) are all present in conserved sequences of RegA. We have recently found that the block to sporulation and aberrant stalk formation resulting from

inactivation of the *dhkA* gene that encodes a histidine kinase can be suppressed by disruption of the *regA* gene (N. Wang, G. Shaulsky and W. F. Loomis, unpublished). Likewise, we found that overexpression of *pkaC* or disruption of *pkaR* can suppress the phenotypic defects resulting from *dhkA*⁻ mutations. These genetic analyses strongly indicate that the signal transduction pathway extends from the DhkA histidine kinase to the response regulator, RegA, to affect the activity of PKA. In a linear pathway, activation of DhkA would lead to inhibition of RegA activity, resulting in accumulation of cAMP-bound PKA-R and free catalytic subunits of PKA. The fact that RegA synthesized in bacteria is an active cAMP-PDE, and that it is able to accept a phosphate on an aspartate from acetyl phosphate, further supports that interpretation.

The model that emerges from our findings is presented in Fig. 6. In wild-type prespore cells, pre-mature sporulation is prevented by inhibition of PKA-C activity by PKA-R. The cAMP-PDE activity of RegA degrades cAMP and keeps the activity of PKA low. When optimal conditions for sporulation are reached, prestalk cells emit a signal that is received by prespore cells and the inhibition of PKA-C is lifted. The model predicts that activation of the two-component pathway results in the inhibition of RegA and a subsequent rise in cAMP that triggers encapsulation via PKA. Stimulation of RegA PDE activity by interaction with PKA-R and inhibition of adenylyl cyclase by PKA (Abe and Yanagisawa, 1983) indicate that the system involves several feedback loops. Both constitutive and reduced PKA activity result in poor encapsulation.

Proper tuning of PKA-C activity appears to be essential in other organisms as well. Female reproduction as well as learning and memory in both sexes of the fruit fly *D. melanogaster* are affected by mutations in the adenylyl cyclase gene *rutabaga*, the phosphodiesterase gene, *dunce*, and the PKA-C gene, *DCO*, suggesting that either an increase or a decrease in PKA activity is deleterious to these processes (Bellen et al., 1987; Bellen and Kiger, 1987, 1988; Davis, 1993). All of these genes are preferentially expressed in the mushroom body of the fly brain, where they may interact (Davis, 1993; Skoulakis et al., 1993). In vertebrates, activation of PKA appears to play a major role in long-term potentiation when neurotransmitters stimulate adenylyl cyclase activity (Abel et al., 1997). Return to basal levels of PKA activity would be expected to require clearing of cAMP from the regulatory subunit by a phosphodiesterase. It is not known which of the many different PKA-R isoforms and cyclic nucleotide phosphodiesterases function in vertebrate nervous systems (Beavo, 1990; Taylor et al., 1990). However, it was a dominant negative form of PKA-R^I_a that affected long-term memory when expressed in mouse brain and that is the isoform that stimulates RegA-PDE activity (Fig. 3). A mammalian phosphodiesterase similar to *dunce* has been found to be highly expressed in rat brain (Davis et al., 1989) and so is a good candidate for direct interaction with PKA-R. The conservation of the circuitry components in these diverse organisms suggests that interactions between PKA-R and cAMP phosphodiesterases may be widespread.

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